

Volume Ii The Continuum Reference

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Rational Numbers ($\mathbb{Q}$)

0.1 Construction of $\mathbb{Q}$

Prefractions

Definition 0.1.1 (Nonzero Integers). The set of nonzero integers is

$$\mathbb{Z}^* := \mathbb{Z} \setminus \{0_{\mathbb{Z}}\}.$$

Definition 0.1.2 (Prefraction). A *prefraction* is an ordered pair $(a, b) \in \mathbb{Z} \times \mathbb{Z}^*$, read as the formal quotient a/b .

Definition 0.1.3 (Prefraction Set). The prefraction set is

$$\text{Pre } \mathbb{Q} := \mathbb{Z} \times \mathbb{Z}^*.$$

Fraction Equivalence

Definition 0.1.4 (Fraction Equivalence). For $(a, b), (c, d) \in \text{Pre } \mathbb{Q}$, define

$$(a, b) \sim_{\mathbb{Q}} (c, d) \quad :\Longleftrightarrow \quad a \cdot d = b \cdot c.$$

Lemma 0.1.5 (Fraction Equivalence Is Reflexive). *For every $(a, b) \in \text{Pre } \mathbb{Q}$,*

$$(a, b) \sim_{\mathbb{Q}} (a, b).$$

Lemma 0.1.6 (Fraction Equivalence Is Symmetric). *For all prefractions $(a, b), (c, d)$,*

$$(a, b) \sim_{\mathbb{Q}} (c, d) \implies (c, d) \sim_{\mathbb{Q}} (a, b).$$

Lemma 0.1.7 (Fraction Equivalence Is Transitive). *For all prefractions $(a, b), (c, d), (e, f)$,*

$$(a, b) \sim_{\mathbb{Q}} (c, d) \wedge (c, d) \sim_{\mathbb{Q}} (e, f) \implies (a, b) \sim_{\mathbb{Q}} (e, f).$$

Theorem 0.1.8 (Fraction Equivalence Is an Equivalence Relation). *The relation $\sim_{\mathbb{Q}}$ is an equivalence relation on $\text{Pre } \mathbb{Q}$.*

Definition of $\mathbb{Q}$

Definition 0.1.9 (Rational Numbers). The rational numbers are the quotient

$$\mathbb{Q} := \text{Pre } \mathbb{Q} / \sim_{\mathbb{Q}}.$$

Definition 0.1.10 (Rational Class Notation). The notation $[a, b]$ denotes the $\sim_{\mathbb{Q}}$ -equivalence class of the prefraction (a, b) , where $b \neq 0_{\mathbb{Z}}$.

Proposition 0.1.11 (Equality Criterion for Rational Classes). *For prefractions $(a, b), (c, d) \in \text{Pre } \mathbb{Q}$,*

$$[a, b] = [c, d] \quad \Longleftrightarrow \quad a \cdot d = b \cdot c.$$

Zero and One in $\mathbb{Q}$

Definition 0.1.12 (Zero in $\mathbb{Q}$). Define

$$0_{\mathbb{Q}} := [0_{\mathbb{Z}}, 1_{\mathbb{Z}}].$$

Definition 0.1.13 (One in $\mathbb{Q}$). Define

$$1_{\mathbb{Q}} := [1_{\mathbb{Z}}, 1_{\mathbb{Z}}].$$

Lemma 0.1.14 (Vanishing Criterion in $\mathbb{Q}$). *For every prefraction $(a, b) \in \text{Pre } \mathbb{Q}$,*

$$[a, b] = 0_{\mathbb{Q}} \iff a = 0_{\mathbb{Z}}.$$

Theorem 0.1.15 (Zero Is a Rational). *The class $0_{\mathbb{Q}}$ is an element of $\mathbb{Q}$.*

Theorem 0.1.16 (One Is a Rational). *The class $1_{\mathbb{Q}}$ is an element of $\mathbb{Q}$.*

Theorem 0.1.17 (Zero Is Not One in $\mathbb{Q}$). *In $\mathbb{Q}$,*

$$0_{\mathbb{Q}} \neq 1_{\mathbb{Q}}.$$

0.2 Operations on $\mathbb{Q}$ -Classes

Addition on $\mathbb{Q}$ -Classes

Definition 0.2.1 (Addition on $\mathbb{Q}$ -Classes). Define addition on representatives by

$$[a, b] + [c, d] := [a \cdot d + b \cdot c, b \cdot d].$$

Lemma 0.2.2 (Addition Lands in $\text{Pre } \mathbb{Q}$). *If $b \neq 0_{\mathbb{Z}}$ and $d \neq 0_{\mathbb{Z}}$, then*

$$b \cdot d \neq 0_{\mathbb{Z}}.$$

Lemma 0.2.3 (Addition Respects Fraction Equivalence on the Left). *If $[a, b] = [a', b']$, then*

$$[a, b] + [c, d] = [a', b'] + [c, d].$$

Lemma 0.2.4 (Addition Respects Fraction Equivalence on the Right). *If $[c, d] = [c', d']$, then*

$$[a, b] + [c, d] = [a, b] + [c', d'].$$

Lemma 0.2.5 (Addition Respects Fraction Equivalence). *If $[a, b] = [a', b']$ and $[c, d] = [c', d']$, then*

$$[a, b] + [c, d] = [a', b'] + [c', d'].$$

Theorem 0.2.6 (Addition on $\mathbb{Q}$ Is Well-Defined). *Addition determines a well-defined binary operation $\mathbb{Q} \times \mathbb{Q} \rightarrow \mathbb{Q}$.*

Theorem 0.2.7 (Addition on $\mathbb{Q}$ Is Associative). *For all $x, y, z \in \mathbb{Q}$,*

$$(x + y) + z = x + (y + z).$$

Theorem 0.2.8 (Addition on $\mathbb{Q}$ Is Commutative). *For all $x, y \in \mathbb{Q}$,*

$$x + y = y + x.$$

Theorem 0.2.9 (Zero Is a Left Additive Identity on $\mathbb{Q}$). *For every $x \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} + x = x.$$

Corollary 0.2.10 (Zero Is a Right Additive Identity on $\mathbb{Q}$). *For every $x \in \mathbb{Q}$,*

$$x + 0_{\mathbb{Q}} = x.$$

Theorem 0.2.11 (Zero Is an Additive Identity on $\mathbb{Q}$). *For every $x \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} + x = x \quad \text{and} \quad x + 0_{\mathbb{Q}} = x.$$

Negation on $\mathbb{Q}$ -Classes

Definition 0.2.12 (Negation on $\mathbb{Q}$ -Classes). Define negation on representatives by

$$-[a, b] := [-a, b].$$

Lemma 0.2.13 (Negation Respects Fraction Equivalence). *If $[a, b] = [a', b']$, then*

$$-[a, b] = -[a', b'].$$

Theorem 0.2.14 (Negation on $\mathbb{Q}$ Is Well-Defined). *Negation determines a well-defined unary operation $\mathbb{Q} \rightarrow \mathbb{Q}$.*

Theorem 0.2.15 (Negation Gives a Left Additive Inverse on $\mathbb{Q}$). *For every $x \in \mathbb{Q}$,*

$$(-x) + x = 0_{\mathbb{Q}}.$$

Corollary 0.2.16 (Negation Gives a Right Additive Inverse on $\mathbb{Q}$). *For every $x \in \mathbb{Q}$,*

$$x + (-x) = 0_{\mathbb{Q}}.$$

Theorem 0.2.17 (Every Rational Has an Additive Inverse). *For every $x \in \mathbb{Q}$ there exists $y \in \mathbb{Q}$ such that*

$$x + y = 0_{\mathbb{Q}} \quad \text{and} \quad y + x = 0_{\mathbb{Q}}.$$

Subtraction on $\mathbb{Q}$

Definition 0.2.18 (Subtraction on $\mathbb{Q}$). For $x, y \in \mathbb{Q}$, define

$$x - y := x + (-y).$$

Theorem 0.2.19 (Subtraction on $\mathbb{Q}$ Is Well-Defined). *Subtraction determines a well-defined binary operation $\mathbb{Q} \times \mathbb{Q} \rightarrow \mathbb{Q}$.*

Multiplication on $\mathbb{Q}$ -Classes

Definition 0.2.20 (Multiplication on $\mathbb{Q}$ -Classes). Define multiplication on representatives by

$$[a, b] \cdot [c, d] := [a \cdot c, b \cdot d].$$

Lemma 0.2.21 (Multiplication Lands in $\text{Pre } \mathbb{Q}$). *If $b \neq 0_{\mathbb{Z}}$ and $d \neq 0_{\mathbb{Z}}$, then*

$$b \cdot d \neq 0_{\mathbb{Z}}.$$

Lemma 0.2.22 (Multiplication Respects Fraction Equivalence on the Left). *If $[a, b] = [a', b']$, then*

$$[a, b] \cdot [c, d] = [a', b'] \cdot [c, d].$$

Lemma 0.2.23 (Multiplication Respects Fraction Equivalence on the Right). *If $[c, d] = [c', d']$, then*

$$[a, b] \cdot [c, d] = [a, b] \cdot [c', d'].$$

Lemma 0.2.24 (Multiplication Respects Fraction Equivalence). *If $[a, b] = [a', b']$ and $[c, d] = [c', d']$, then*

$$[a, b] \cdot [c, d] = [a', b'] \cdot [c', d'].$$

Theorem 0.2.25 (Multiplication on $\mathbb{Q}$ Is Well-Defined). *Multiplication determines a well-defined binary operation $\mathbb{Q} \times \mathbb{Q} \rightarrow \mathbb{Q}$.*

Theorem 0.2.26 (Multiplication on $\mathbb{Q}$ Is Associative). *For all $x, y, z \in \mathbb{Q}$,*

$$(x \cdot y) \cdot z = x \cdot (y \cdot z).$$

Theorem 0.2.27 (Multiplication on $\mathbb{Q}$ Is Commutative). *For all $x, y \in \mathbb{Q}$,*

$$x \cdot y = y \cdot x.$$

Theorem 0.2.28 (One Is a Left Multiplicative Identity on $\mathbb{Q}$). *For every $x \in \mathbb{Q}$,*

$$1_{\mathbb{Q}} \cdot x = x.$$

Corollary 0.2.29 (One Is a Right Multiplicative Identity on $\mathbb{Q}$). *For every $x \in \mathbb{Q}$,*

$$x \cdot 1_{\mathbb{Q}} = x.$$

Theorem 0.2.30 (One Is a Multiplicative Identity on $\mathbb{Q}$). *For every $x \in \mathbb{Q}$,*

$$1_{\mathbb{Q}} \cdot x = x \quad \text{and} \quad x \cdot 1_{\mathbb{Q}} = x.$$

Reciprocal on Nonzero $\mathbb{Q}$ -Classes

Lemma 0.2.31 (Nonzero Predicate Is Well-Defined on $\mathbb{Q}$). *For every prefraction $(a, b) \in \text{Pre } \mathbb{Q}$,*

$$[a, b] \neq 0_{\mathbb{Q}} \iff a \neq 0_{\mathbb{Z}}.$$

Thus nonzeroness is independent of the representative.

Definition 0.2.32 (Reciprocal on Nonzero $\mathbb{Q}$ -Classes). For $[a, b] \neq 0_{\mathbb{Q}}$, define

$$[a, b]^{-1} := [b, a].$$

Lemma 0.2.33 (Reciprocal Lands in $\text{Pre } \mathbb{Q}$). *If $[a, b] \neq 0_{\mathbb{Q}}$, then $a \neq 0_{\mathbb{Z}}$, so (b, a) is a prefraction.*

Lemma 0.2.34 (Reciprocal Respects Fraction Equivalence). *If $[a, b] = [a', b']$ and $[a, b] \neq 0_{\mathbb{Q}}$, then*

$$[a, b]^{-1} = [a', b']^{-1}.$$

Theorem 0.2.35 (Reciprocal on $\mathbb{Q}$ Is Well-Defined). *Reciprocal determines a well-defined map*

$$\mathbb{Q} \setminus \{0_{\mathbb{Q}}\} \rightarrow \mathbb{Q} \setminus \{0_{\mathbb{Q}}\}.$$

Theorem 0.2.36 (Reciprocal Gives a Left Multiplicative Inverse on $\mathbb{Q}$). *For every nonzero $x \in \mathbb{Q}$,*

$$x^{-1} \cdot x = 1_{\mathbb{Q}}.$$

Corollary 0.2.37 (Reciprocal Gives a Right Multiplicative Inverse on $\mathbb{Q}$). *For every nonzero $x \in \mathbb{Q}$,*

$$x \cdot x^{-1} = 1_{\mathbb{Q}}.$$

Theorem 0.2.38 (Every Nonzero Rational Has a Multiplicative Inverse). *For every $x \in \mathbb{Q}$ with $x \neq 0_{\mathbb{Q}}$, there exists $y \in \mathbb{Q}$ such that*

$$x \cdot y = 1_{\mathbb{Q}} \quad \text{and} \quad y \cdot x = 1_{\mathbb{Q}}.$$

Division on $\mathbb{Q}$

Definition 0.2.39 (Division on $\mathbb{Q}$). For $x, y \in \mathbb{Q}$ with $y \neq 0_{\mathbb{Q}}$, define

$$x/y := x \cdot y^{-1}.$$

Theorem 0.2.40 (Division on $\mathbb{Q}$ Is Well-Defined). *Division determines a well-defined map*

$$\mathbb{Q} \times (\mathbb{Q} \setminus \{0_{\mathbb{Q}}\}) \rightarrow \mathbb{Q}.$$

Distributivity on $\mathbb{Q}$ -Classes

Theorem 0.2.41 (Left Distributivity on $\mathbb{Q}$). *For all $x, y, z \in \mathbb{Q}$,*

$$x \cdot (y + z) = x \cdot y + x \cdot z.$$

Corollary 0.2.42 (Right Distributivity on $\mathbb{Q}$). *For all $x, y, z \in \mathbb{Q}$,*

$$(y + z) \cdot x = y \cdot x + z \cdot x.$$

Theorem 0.2.43 (Multiplication Distributes over Addition on $\mathbb{Q}$). *Multiplication distributes over addition on both sides in $\mathbb{Q}$.*

Class-Level Field Summary

Theorem 0.2.44 ($\mathbb{Q}$ Is an Additive Abelian Group). *The structure $(\mathbb{Q}, +, 0_{\mathbb{Q}})$ is an abelian group.*

Theorem 0.2.45 (Nonzero $\mathbb{Q}$ Is a Multiplicative Abelian Group). *The structure $(\mathbb{Q} \setminus \{0_{\mathbb{Q}}\}, \cdot, 1_{\mathbb{Q}})$ is an abelian group.*

Theorem 0.2.46 ($\mathbb{Q}$ Is a Field). *The structure $(\mathbb{Q}, +, \cdot, 0_{\mathbb{Q}}, 1_{\mathbb{Q}})$ is a field.*

0.3 Field Structure of $\mathbb{Q}$

Derived Field Laws on $\mathbb{Q}$

Corollary 0.3.1 (Zero Product on $\mathbb{Q}$). *For all $x, y \in \mathbb{Q}$,*

$$x \cdot y = 0_{\mathbb{Q}} \iff x = 0_{\mathbb{Q}} \text{ or } y = 0_{\mathbb{Q}}.$$

Corollary 0.3.2 (Double Negation on $\mathbb{Q}$). *For every $x \in \mathbb{Q}$,*

$$-(-x) = x.$$

Corollary 0.3.3 (Negation of a Product on $\mathbb{Q}$). *For all $x, y \in \mathbb{Q}$,*

$$(-x) \cdot y = -(x \cdot y) \quad \text{and} \quad x \cdot (-y) = -(x \cdot y).$$

Corollary 0.3.4 (Product of Negatives on $\mathbb{Q}$). *For all $x, y \in \mathbb{Q}$,*

$$(-x) \cdot (-y) = x \cdot y.$$

Corollary 0.3.5 (Additive Cancellation on $\mathbb{Q}$). *For all $x, y, z \in \mathbb{Q}$,*

$$x + z = y + z \implies x = y.$$

Corollary 0.3.6 (Multiplicative Cancellation on $\mathbb{Q}$). *For all $x, y, z \in \mathbb{Q}$,*

$$z \neq 0_{\mathbb{Q}} \wedge x \cdot z = y \cdot z \implies x = y.$$

Corollary 0.3.7 (Uniqueness of Additive Inverses on $\mathbb{Q}$). *Every additive inverse in $\mathbb{Q}$ is unique.*

Corollary 0.3.8 (Uniqueness of Multiplicative Inverses on $\mathbb{Q}$). *Every multiplicative inverse of a nonzero rational number is unique.*

Corollary 0.3.9 (The Inverse of One Is One on $\mathbb{Q}$). *In $\mathbb{Q}$,*

$$1_{\mathbb{Q}}^{-1} = 1_{\mathbb{Q}}.$$

Corollary 0.3.10 (Inverse of a Product on $\mathbb{Q}$). *For all nonzero $x, y \in \mathbb{Q}$,*

$$(x \cdot y)^{-1} = x^{-1} \cdot y^{-1}.$$

Corollary 0.3.11 (Inverse of an Inverse on $\mathbb{Q}$). *For every nonzero $x \in \mathbb{Q}$,*

$$(x^{-1})^{-1} = x.$$

Integer Powers on $\mathbb{Q}$

Definition 0.3.12 (Integer Power on $\mathbb{Q}$). Let $x \in \mathbb{Q}$. For $n \in \mathbb{W}$, define x^n by the usual recursive power operation. If $x \neq 0_{\mathbb{Q}}$ and $n \in \mathbb{Z}$ is negative, define

$$x^n := (x^{-1})^{-n}.$$

Corollary 0.3.13 (Exponent Addition Law on $\mathbb{Q}$). For every nonzero $x \in \mathbb{Q}$ and all $m, n \in \mathbb{Z}$,

$$x^{m+n} = x^m \cdot x^n.$$

Corollary 0.3.14 (Power of a Power on $\mathbb{Q}$). For every nonzero $x \in \mathbb{Q}$ and all $m, n \in \mathbb{Z}$,

$$(x^m)^n = x^{m \cdot n}.$$

Corollary 0.3.15 (Power of a Product on $\mathbb{Q}$). For all nonzero $x, y \in \mathbb{Q}$ and every $n \in \mathbb{Z}$,

$$(x \cdot y)^n = x^n \cdot y^n.$$

0.4 Order on $\mathbb{Q}$

Positivity on $\mathbb{Q}$

Definition 0.4.1 (Positive Rational). A rational number $[a, b] \in \mathbb{Q}$ is *positive* if

$$a \cdot b > 0_{\mathbb{Z}}.$$

Lemma 0.4.2 (Positivity Respects Fraction Equivalence on $\mathbb{Q}$). If $[a, b] = [c, d]$, then

$$a \cdot b > 0_{\mathbb{Z}} \iff c \cdot d > 0_{\mathbb{Z}}.$$

Theorem 0.4.3 (Positivity on $\mathbb{Q}$ Is Well-Defined). The predicate “ x is positive” is well-defined on equivalence classes in $\mathbb{Q}$.

Definition 0.4.4 (Strict Order on $\mathbb{Q}$). For $x, y \in \mathbb{Q}$, define

$$x < y \iff y - x \text{ is positive.}$$

Definition 0.4.5 (Order on $\mathbb{Q}$). For $x, y \in \mathbb{Q}$, define

$$x \leq y \iff x < y \text{ or } x = y.$$

Proposition 0.4.6 (Product of Positives Is Positive on $\mathbb{Q}$). For all $x, y \in \mathbb{Q}$,

$$0_{\mathbb{Q}} < x \wedge 0_{\mathbb{Q}} < y \implies 0_{\mathbb{Q}} < x \cdot y.$$

Trichotomy on $\mathbb{Q}$

Theorem 0.4.7 (Sign Trichotomy on $\mathbb{Q}$). For every $x \in \mathbb{Q}$, exactly one of

$$x < 0_{\mathbb{Q}}, \quad x = 0_{\mathbb{Q}}, \quad 0_{\mathbb{Q}} < x$$

holds.

Theorem 0.4.8 (Trichotomy on $\mathbb{Q}$). For all $x, y \in \mathbb{Q}$, exactly one of

$$x < y, \quad x = y, \quad y < x$$

holds.

Total Order on $\mathbb{Q}$

Theorem 0.4.9 (Strict Order on $\mathbb{Q}$ Is Irreflexive). *For every $x \in \mathbb{Q}$, not $x < x$.*

Theorem 0.4.10 (Strict Order on $\mathbb{Q}$ Is Asymmetric). *For all $x, y \in \mathbb{Q}$,*

$$x < y \implies \neg(y < x).$$

Theorem 0.4.11 (Strict Order on $\mathbb{Q}$ Is Transitive). *For all $x, y, z \in \mathbb{Q}$,*

$$x < y \wedge y < z \implies x < z.$$

Theorem 0.4.12 (Order on $\mathbb{Q}$ Is Reflexive). *For every $x \in \mathbb{Q}$, $x \leq x$.*

Theorem 0.4.13 (Order on $\mathbb{Q}$ Is Antisymmetric). *For all $x, y \in \mathbb{Q}$,*

$$x \leq y \wedge y \leq x \implies x = y.$$

Theorem 0.4.14 (Order on $\mathbb{Q}$ Is Transitive). *For all $x, y, z \in \mathbb{Q}$,*

$$x \leq y \wedge y \leq z \implies x \leq z.$$

Theorem 0.4.15 (Order on $\mathbb{Q}$ Is Total). *For all $x, y \in \mathbb{Q}$,*

$$x \leq y \text{ or } y \leq x.$$

Theorem 0.4.16 ($\mathbb{Q}$ Is a Totally Ordered Set). *The ordered pair $(\mathbb{Q}, \leq)$ is a totally ordered set.*

Addition Order Compatibility on $\mathbb{Q}$

Theorem 0.4.17 (Left Addition Preserves Strict Order on $\mathbb{Q}$). *For all $x, y, z \in \mathbb{Q}$,*

$$x < y \implies z + x < z + y.$$

Corollary 0.4.18 (Right Addition Preserves Strict Order on $\mathbb{Q}$). *For all $x, y, z \in \mathbb{Q}$,*

$$x < y \implies x + z < y + z.$$

Theorem 0.4.19 (Addition Preserves Strict Order on $\mathbb{Q}$). *Addition preserves strict order on both sides in $\mathbb{Q}$.*

Theorem 0.4.20 (Left Addition Preserves Order on $\mathbb{Q}$). *For all $x, y, z \in \mathbb{Q}$,*

$$x \leq y \implies z + x \leq z + y.$$

Corollary 0.4.21 (Right Addition Preserves Order on $\mathbb{Q}$). *For all $x, y, z \in \mathbb{Q}$,*

$$x \leq y \implies x + z \leq y + z.$$

Theorem 0.4.22 (Addition Preserves Order on $\mathbb{Q}$). *Addition preserves non-strict order on both sides in $\mathbb{Q}$.*

Corollary 0.4.23 (Addition Reflects Order on $\mathbb{Q}$). *Addition by a fixed rational reflects both strict and non-strict order.*

Multiplication Order Compatibility on $\mathbb{Q}$

Theorem 0.4.24 (Left Multiplication by Positives Preserves Strict Order on $\mathbb{Q}$). *For all $x, y, z \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} < z \wedge x < y \implies z \cdot x < z \cdot y.$$

Corollary 0.4.25 (Right Multiplication by Positives Preserves Strict Order on $\mathbb{Q}$). *For all $x, y, z \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} < z \wedge x < y \implies x \cdot z < y \cdot z.$$

Theorem 0.4.26 (Multiplication by Positives Preserves Strict Order on $\mathbb{Q}$). *Multiplication by a positive rational preserves strict order on both sides.*

Theorem 0.4.27 (Left Multiplication by Positives Preserves Order on $\mathbb{Q}$). *For all $x, y, z \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} < z \wedge x \leq y \implies z \cdot x \leq z \cdot y.$$

Corollary 0.4.28 (Right Multiplication by Positives Preserves Order on $\mathbb{Q}$). *For all $x, y, z \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} < z \wedge x \leq y \implies x \cdot z \leq y \cdot z.$$

Theorem 0.4.29 (Multiplication by Positives Preserves Order on $\mathbb{Q}$). *Multiplication by a positive rational preserves non-strict order on both sides.*

Theorem 0.4.30 (Multiplication by Negatives Reverses Strict Order on $\mathbb{Q}$). *For all $x, y, z \in \mathbb{Q}$,*

$$z < 0_{\mathbb{Q}} \wedge x < y \implies z \cdot y < z \cdot x.$$

Corollary 0.4.31 (Multiplication by Negatives Reverses Order on $\mathbb{Q}$). *For all $x, y, z \in \mathbb{Q}$,*

$$z < 0_{\mathbb{Q}} \wedge x \leq y \implies z \cdot y \geq z \cdot x.$$

Reciprocal Order Laws on $\mathbb{Q}$

Theorem 0.4.32 (Sign of a Reciprocal on $\mathbb{Q}$). *For every nonzero $x \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} < x \implies 0_{\mathbb{Q}} < x^{-1}, \quad x < 0_{\mathbb{Q}} \implies x^{-1} < 0_{\mathbb{Q}}.$$

Theorem 0.4.33 (Reciprocal Reverses Order on Positives in $\mathbb{Q}$). *For all $x, y \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} < x < y \implies 0_{\mathbb{Q}} < y^{-1} < x^{-1}.$$

Absolute Value on $\mathbb{Q}$

Definition 0.4.34 (Absolute Value on $\mathbb{Q}$). For $x \in \mathbb{Q}$, define

$$|x| := \begin{cases} x, & 0_{\mathbb{Q}} \leq x, \\ -x, & x < 0_{\mathbb{Q}}. \end{cases}$$

Theorem 0.4.35 (Absolute Value Is Nonnegative on $\mathbb{Q}$). *For every $x \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} \leq |x|.$$

Theorem 0.4.36 (Absolute Value Vanishes Only at Zero on $\mathbb{Q}$). *For every $x \in \mathbb{Q}$,*

$$|x| = 0_{\mathbb{Q}} \iff x = 0_{\mathbb{Q}}.$$

Theorem 0.4.37 (Absolute Value Is Multiplicative on $\mathbb{Q}$). *For all $x, y \in \mathbb{Q}$,*

$$|x \cdot y| = |x| \cdot |y|.$$

Theorem 0.4.38 (Triangle Inequality on $\mathbb{Q}$). *For all $x, y \in \mathbb{Q}$,*

$$|x + y| \leq |x| + |y|.$$

Theorem 0.4.39 (Absolute Value of a Reciprocal on $\mathbb{Q}$). *For every nonzero $x \in \mathbb{Q}$,*

$$|x^{-1}| = |x|^{-1}.$$

0.5 Ordered Field Structure of $\mathbb{Q}$

Ordered Field Instantiation

Theorem 0.5.1 ($\mathbb{Q}$ Is an Ordered Field). *The structure*

$$(\mathbb{Q}, +, \cdot, 0_{\mathbb{Q}}, 1_{\mathbb{Q}}, \leq)$$

is an ordered field.

Corollary 0.5.2 (Ordered Field Laws Hold on $\mathbb{Q}$). *Every theorem that depends only on the ordered-field axioms applies to $\mathbb{Q}$ with its constructed operations and order.*

Standard Ordered-Field Consequences

Corollary 0.5.3 (Squares Are Nonnegative on $\mathbb{Q}$). *For every $x \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} \leq x^2.$$

Corollary 0.5.4 (One Is Positive in $\mathbb{Q}$). *In $\mathbb{Q}$,*

$$0_{\mathbb{Q}} < 1_{\mathbb{Q}}.$$

Corollary 0.5.5 (Natural Numbers Are Positive in $\mathbb{Q}$). *Every natural number embedded in $\mathbb{Q}$ is positive.*

Corollary 0.5.6 (Positive Rationals Have Positive Inverses). *For every $x \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} < x \implies 0_{\mathbb{Q}} < x^{-1}.$$

Corollary 0.5.7 (Fraction Comparison in $\mathbb{Q}$). *For rational classes with positive denominator product,*

$$[a, b] < [c, d] \iff a \cdot d < c \cdot b.$$

0.6 Arbitrary Closeness in $\mathbb{Q}$

Density and the Archimedean Property

Definition 0.6.1 (Two in $\mathbb{Q}$). Define

$$2_{\mathbb{Q}} := 1_{\mathbb{Q}} + 1_{\mathbb{Q}}.$$

Lemma 0.6.2 (Two Is Nonzero in $\mathbb{Q}$). *In $\mathbb{Q}$,*

$$2_{\mathbb{Q}} \neq 0_{\mathbb{Q}}.$$

Lemma 0.6.3 (Midpoint Lies Strictly Between Rationals). *For all $x, y \in \mathbb{Q}$,*

$$x < y \implies x < (x + y) \cdot 2_{\mathbb{Q}}^{-1} < y.$$

Theorem 0.6.4 (Density of $\mathbb{Q}$ in Itself). *For all $x, y \in \mathbb{Q}$,*

$$x < y \implies \exists r \in \mathbb{Q} (x < r < y).$$

Theorem 0.6.5 (Archimedean Property of $\mathbb{Q}$). *For all $x, y \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} < x \implies \exists n \in \mathbb{N} (y < n \cdot x).$$

From Discreteness to Arbitrary Closeness

Definition 0.6.6 (Arbitrarily Close Distinct Points). A number system has arbitrarily close distinct points if for every point x and every positive tolerance ε , there is a point $y \neq x$ such that

$$|x - y| < \varepsilon.$$

Theorem 0.6.7 ($\mathbb{Q}$ Has Arbitrarily Close Distinct Points). For every $x \in \mathbb{Q}$ and every $\varepsilon \in \mathbb{Q}$,

$$0_{\mathbb{Q}} < \varepsilon \implies \exists y \in \mathbb{Q} (y \neq x \text{ and } |x - y| < \varepsilon).$$

Corollary 0.6.8 ($\mathbb{Q}$ Has No Adjacent Points). There do not exist rational numbers $x < y$ with no rational number strictly between them.

Proposition 0.6.9 ($\mathbb{Q}$ Has No Least Positive Rational). For every $x \in \mathbb{Q}$,

$$0_{\mathbb{Q}} < x \implies \exists y \in \mathbb{Q} (0_{\mathbb{Q}} < y < x).$$

Corollary 0.6.10 (Reciprocals Become Arbitrarily Small in $\mathbb{Q}$). For every $\varepsilon \in \mathbb{Q}$,

$$0_{\mathbb{Q}} < \varepsilon \implies \exists n \in \mathbb{N} (0_{\mathbb{Q}} < n^{-1} < \varepsilon).$$

Limit-Like Behavior Before Limits

Proposition 0.6.11 (One-Sided Rational Approximation). For every $x \in \mathbb{Q}$ and every $\varepsilon \in \mathbb{Q}$,

$$0_{\mathbb{Q}} < \varepsilon \implies \exists y, z \in \mathbb{Q} (x - \varepsilon < y < x < z < x + \varepsilon).$$

0.7 Rational Gaps

No Rational Square Root of Two

Theorem 0.7.1 (No Rational Square Root of Two). There is no rational number $q \in \mathbb{Q}$ such that

$$q \cdot q = 2_{\mathbb{Q}}.$$

The Square-Two Gap

Definition 0.7.2 (The Gap Set in $\mathbb{Q}$). Define

$$S := \{q \in \mathbb{Q} : 0_{\mathbb{Q}} < q \text{ and } q \cdot q < 2_{\mathbb{Q}}\}.$$

Proposition 0.7.3 (The Gap Set Is Bounded Above in $\mathbb{Q}$). The set S is nonempty and bounded above in $\mathbb{Q}$.

Theorem 0.7.4 (The Square-Two Gap Has No Rational Supremum). The set S has no least upper bound in $\mathbb{Q}$.

Corollary 0.7.5 ($\mathbb{Q}$ Is Order-Incomplete). The ordered field $\mathbb{Q}$ is not order-complete: there exists a nonempty bounded-above subset of $\mathbb{Q}$ with no supremum in $\mathbb{Q}$.

Cauchy Sequences in $\mathbb{Q}$

Definition 0.7.6 (Cauchy Sequence in $\mathbb{Q}$). A sequence (x_n) in $\mathbb{Q}$ is Cauchy if

$$\forall \varepsilon \in \mathbb{Q} (0_{\mathbb{Q}} < \varepsilon \implies \exists N \in \mathbb{N} \forall m, n \geq N |x_m - x_n| < \varepsilon).$$

Definition 0.7.7 (Null Sequence in $\mathbb{Q}$). A sequence (x_n) in $\mathbb{Q}$ is null if

$$\forall \varepsilon \in \mathbb{Q} (0_{\mathbb{Q}} < \varepsilon \implies \exists N \in \mathbb{N} \forall n \geq N |x_n| < \varepsilon).$$

Lemma 0.7.8 (Rational Limits Are Unique). A convergent sequence in $\mathbb{Q}$ has at most one rational limit.

0.8 Embedding $\mathbb{Z}$ into $\mathbb{Q}$

The Integer Embedding

Definition 0.8.1 (Embedding of $\mathbb{Z}$ into $\mathbb{Q}$). Define

$$\iota : \mathbb{Z} \rightarrow \mathbb{Q}, \quad \iota(n) := [n, 1_{\mathbb{Z}}].$$

Theorem 0.8.2 (Embedding Preserves Zero). *The embedding sends integer zero to rational zero:*

$$\iota(0_{\mathbb{Z}}) = 0_{\mathbb{Q}}.$$

Theorem 0.8.3 (Embedding Preserves One). *The embedding sends integer one to rational one:*

$$\iota(1_{\mathbb{Z}}) = 1_{\mathbb{Q}}.$$

Theorem 0.8.4 (Embedding $\mathbb{Z}$ into $\mathbb{Q}$ Is Injective). *For all $m, n \in \mathbb{Z}$,*

$$\iota(m) = \iota(n) \implies m = n.$$

Preservation of Algebra and Order

Theorem 0.8.5 (Embedding Preserves Addition). *For all $m, n \in \mathbb{Z}$,*

$$\iota(m + n) = \iota(m) + \iota(n).$$

Theorem 0.8.6 (Embedding Preserves Multiplication). *For all $m, n \in \mathbb{Z}$,*

$$\iota(m \cdot n) = \iota(m) \cdot \iota(n).$$

Theorem 0.8.7 (Embedding Preserves Order). *For all $m, n \in \mathbb{Z}$,*

$$m \leq n \iff \iota(m) \leq \iota(n).$$

Theorem 0.8.8 ($\mathbb{Z}$ Embeds into $\mathbb{Q}$). *The map $\iota : \mathbb{Z} \rightarrow \mathbb{Q}$ is an injective order-preserving ring homomorphism.*

Field of Fractions

Theorem 0.8.9 ($\mathbb{Q}$ Is the Field of Fractions of $\mathbb{Z}$). *For every $x \in \mathbb{Q}$, there exist $m, n \in \mathbb{Z}$ with $n \neq 0_{\mathbb{Z}}$ such that*

$$x = \iota(m) \cdot \iota(n)^{-1}.$$

Corollary 0.8.10 (The Embedding of $\mathbb{Z}$ into $\mathbb{Q}$ Is Not Surjective). *The image of $\iota : \mathbb{Z} \rightarrow \mathbb{Q}$ is a proper subset of $\mathbb{Q}$.*

0.9 Transition to $\mathbb{R}$

Why the Real Numbers Are Introduced

0.10 Canonical Rational Number Statement Plan

Proofs

Real Numbers ($\mathbb{R}$)

0.11 Construction of $\mathbb{R}$

Dedekind Cuts

Definition 0.11.1 (Dedekind Cut). A *Dedekind cut* is a set $\alpha \subseteq \mathbb{Q}$ satisfying:

$$\alpha \neq \emptyset, \quad \alpha \neq \mathbb{Q},$$

$$p \in \alpha \wedge q < p \implies q \in \alpha,$$

and

$$p \in \alpha \implies \exists r \in \alpha (p < r).$$

Definition 0.11.2 (The Real Numbers). The real numbers are the set

$$\mathbb{R} := \{\alpha : \alpha \text{ is a Dedekind cut in } \mathbb{Q}\}.$$

Rational Cuts

Definition 0.11.3 (Rational Cut). For $q \in \mathbb{Q}$, define the rational cut

$$q^* := \{r \in \mathbb{Q} : r < q\}.$$

Lemma 0.11.4 (Rational Cuts Are Cuts). *For every $q \in \mathbb{Q}$, the set q^* is a Dedekind cut.*

Zero and One in $\mathbb{R}$

Definition 0.11.5 (Zero in $\mathbb{R}$). Define

$$0_{\mathbb{R}} := (0_{\mathbb{Q}})^* = \{r \in \mathbb{Q} : r < 0_{\mathbb{Q}}\}.$$

Definition 0.11.6 (One in $\mathbb{R}$). Define

$$1_{\mathbb{R}} := (1_{\mathbb{Q}})^* = \{r \in \mathbb{Q} : r < 1_{\mathbb{Q}}\}.$$

Theorem 0.11.7 (Zero and One Are Reals). *The cuts $0_{\mathbb{R}}$ and $1_{\mathbb{R}}$ are elements of $\mathbb{R}$.*

Theorem 0.11.8 (Zero Is Not One in $\mathbb{R}$). *In $\mathbb{R}$,*

$$0_{\mathbb{R}} \neq 1_{\mathbb{R}}.$$

0.12 Order on $\mathbb{R}$

Order by Inclusion

Definition 0.12.1 (Order on $\mathbb{R}$). For cuts $\alpha, \beta \in \mathbb{R}$, define

$$\alpha \leq \beta \quad :\Longleftrightarrow \quad \alpha \subseteq \beta.$$

Definition 0.12.2 (Strict Order on $\mathbb{R}$). For cuts $\alpha, \beta \in \mathbb{R}$, define

$$\alpha < \beta \quad :\Longleftrightarrow \quad \alpha \subsetneq \beta.$$

Theorem 0.12.3 (Order on $\mathbb{R}$ Is Reflexive). *For every $\alpha \in \mathbb{R}$, $\alpha \leq \alpha$.*

Theorem 0.12.4 (Order on $\mathbb{R}$ Is Antisymmetric). *For all $\alpha, \beta \in \mathbb{R}$,*

$$\alpha \leq \beta \wedge \beta \leq \alpha \implies \alpha = \beta.$$

Theorem 0.12.5 (Order on $\mathbb{R}$ Is Transitive). *For all $\alpha, \beta, \gamma \in \mathbb{R}$,*

$$\alpha \leq \beta \wedge \beta \leq \gamma \implies \alpha \leq \gamma.$$

Theorem 0.12.6 (Comparability of Cuts). *For all cuts $\alpha, \beta \in \mathbb{R}$,*

$$\alpha \subseteq \beta \vee \beta \subseteq \alpha.$$

Theorem 0.12.7 (Order on $\mathbb{R}$ Is Total). *For all $\alpha, \beta \in \mathbb{R}$,*

$$\alpha \leq \beta \vee \beta \leq \alpha.$$

Theorem 0.12.8 (Trichotomy on $\mathbb{R}$). *For all $\alpha, \beta \in \mathbb{R}$, exactly one of*

$$\alpha < \beta, \quad \alpha = \beta, \quad \beta < \alpha$$

holds.

Theorem 0.12.9 ($\mathbb{R}$ Is a Totally Ordered Set). *The ordered pair $(\mathbb{R}, \leq)$ is a total order.*

0.13 Completeness of $\mathbb{R}$

Definition 0.13.1 (Upper Bound in $\mathbb{R}$). Let $S \subseteq \mathbb{R}$. A cut $\gamma \in \mathbb{R}$ is an upper bound of S if

$$\forall \alpha \in S (\alpha \leq \gamma).$$

The set S is bounded above if it has an upper bound in $\mathbb{R}$.

Definition 0.13.2 (Supremum in $\mathbb{R}$). For $S \subseteq \mathbb{R}$, a cut σ is the supremum of S if σ is an upper bound of S and every upper bound γ of S satisfies $\sigma \leq \gamma$.

Definition 0.13.3 (Least-Upper-Bound Property on $\mathbb{R}$). The ordered set $\mathbb{R}$ has the *least-upper-bound property* if every nonempty subset of $\mathbb{R}$ that is bounded above has a supremum in $\mathbb{R}$.

Lemma 0.13.4 (Union of a Bounded Nonempty Family of Cuts Is a Cut). *If $S \subseteq \mathbb{R}$ is nonempty and bounded above, then*

$$\bigcup_{\alpha \in S} \alpha$$

is a Dedekind cut.

Theorem 0.13.5 (Least-Upper-Bound Property). *Every nonempty subset $S \subseteq \mathbb{R}$ that is bounded above has a least upper bound, namely*

$$\sup S = \bigcup_{\alpha \in S} \alpha.$$

Corollary 0.13.6 (Greatest-Lower-Bound Property). *Every nonempty subset $S \subseteq \mathbb{R}$ that is bounded below has a greatest lower bound.*

0.14 Addition on $\mathbb{R}$

Definition 0.14.1 (Addition of Cuts). For cuts $\alpha, \beta \in \mathbb{R}$, define

$$\alpha + \beta := \{p + q : p \in \alpha, q \in \beta\}.$$

Lemma 0.14.2 (Sum of Cuts Is a Cut). *If $\alpha, \beta \in \mathbb{R}$, then $\alpha + \beta$ is a Dedekind cut.*

Theorem 0.14.3 (Addition on $\mathbb{R}$ Is Associative). *For all $\alpha, \beta, \gamma \in \mathbb{R}$,*

$$(\alpha + \beta) + \gamma = \alpha + (\beta + \gamma).$$

Theorem 0.14.4 (Addition on $\mathbb{R}$ Is Commutative). *For all $\alpha, \beta \in \mathbb{R}$,*

$$\alpha + \beta = \beta + \alpha.$$

Theorem 0.14.5 (Zero Is an Additive Identity on $\mathbb{R}$). *For every $\alpha \in \mathbb{R}$,*

$$\alpha + 0_{\mathbb{R}} = \alpha.$$

Definition 0.14.6 (Negation of a Cut). For $\alpha \in \mathbb{R}$, define

$$-\alpha := \{p \in \mathbb{Q} : \exists r \in \mathbb{Q} (r > 0 \wedge -p - r \notin \alpha)\}.$$

Lemma 0.14.7 (Negation of a Cut Is a Cut). *If $\alpha \in \mathbb{R}$, then $-\alpha$ is a Dedekind cut.*

Theorem 0.14.8 (Negation Is an Additive Inverse on $\mathbb{R}$). *For every $\alpha \in \mathbb{R}$,*

$$\alpha + (-\alpha) = 0_{\mathbb{R}}.$$

Theorem 0.14.9 ($\mathbb{R}$ Is an Additive Abelian Group). *The structure $(\mathbb{R}, +, 0_{\mathbb{R}})$ is an abelian group.*

Definition 0.14.10 (Subtraction on $\mathbb{R}$). For cuts $\alpha, \beta \in \mathbb{R}$, define

$$\alpha - \beta := \alpha + (-\beta).$$

0.15 Multiplication on $\mathbb{R}$

Definition 0.15.1 (Nonnegative Cut). A cut $\alpha \in \mathbb{R}$ is nonnegative if $0_{\mathbb{R}} \leq \alpha$ and positive if $0_{\mathbb{R}} < \alpha$.

Definition 0.15.2 (Product of Nonnegative Cuts). For nonnegative cuts $\alpha, \beta \in \mathbb{R}$, define

$$\alpha \cdot \beta := \{r \in \mathbb{Q} : r < 0_{\mathbb{Q}}\} \cup \{p \cdot q : p \in \alpha, q \in \beta, p \geq 0_{\mathbb{Q}}, q \geq 0_{\mathbb{Q}}\}.$$

Lemma 0.15.3 (Product of Nonnegative Cuts Is a Cut). *If $\alpha, \beta \geq 0_{\mathbb{R}}$, then $\alpha \cdot \beta$ is a Dedekind cut.*

Definition 0.15.4 (Absolute Value on $\mathbb{R}$). For $\alpha \in \mathbb{R}$, define

$$|\alpha| := \begin{cases} \alpha, & 0_{\mathbb{R}} \leq \alpha, \\ -\alpha, & \alpha < 0_{\mathbb{R}}. \end{cases}$$

Theorem 0.15.5 (Triangle Inequality). *For all $\alpha, \beta \in \mathbb{R}$,*

$$|\alpha + \beta| \leq |\alpha| + |\beta|.$$

Go to proof.

Definition 0.15.6 (Multiplication of Cuts). For cuts $\alpha, \beta \in \mathbb{R}$, define $\alpha \cdot \beta$ by sign cases: use $|\alpha| \cdot |\beta|$ when α and β have the same sign, use $-(|\alpha| \cdot |\beta|)$ when they have opposite signs, and use $0_{\mathbb{R}}$ when either factor is $0_{\mathbb{R}}$.

Lemma 0.15.7 (Product of Cuts Is a Cut). *If $\alpha, \beta \in \mathbb{R}$, then $\alpha \cdot \beta \in \mathbb{R}$.*

Theorem 0.15.8 (Multiplication on $\mathbb{R}$ Is Associative). *For all $\alpha, \beta, \gamma \in \mathbb{R}$,*

$$(\alpha \cdot \beta) \cdot \gamma = \alpha \cdot (\beta \cdot \gamma).$$

Theorem 0.15.9 (Multiplication on $\mathbb{R}$ Is Commutative). *For all $\alpha, \beta \in \mathbb{R}$,*

$$\alpha \cdot \beta = \beta \cdot \alpha.$$

Theorem 0.15.10 (One Is a Multiplicative Identity on $\mathbb{R}$). *For every $\alpha \in \mathbb{R}$,*

$$\alpha \cdot 1_{\mathbb{R}} = \alpha.$$

Definition 0.15.11 (Reciprocal of a Positive Cut). For $\alpha > 0_{\mathbb{R}}$, define

$$\alpha^{-1} := \{r \in \mathbb{Q} : r \leq 0_{\mathbb{Q}}\} \cup \{r \in \mathbb{Q} : r > 0_{\mathbb{Q}} \wedge \exists s \in \mathbb{Q} (s > r \wedge s^{-1} \notin \alpha)\}.$$

Definition 0.15.12 (Reciprocal of a Nonzero Cut). For $\alpha \neq 0_{\mathbb{R}}$, define α^{-1} to be $|\alpha|^{-1}$ if $\alpha > 0_{\mathbb{R}}$ and $-(|\alpha|^{-1})$ if $\alpha < 0_{\mathbb{R}}$.

Lemma 0.15.13 (Reciprocal of a Nonzero Cut Is a Cut). *If $\alpha \neq 0_{\mathbb{R}}$, then $\alpha^{-1} \in \mathbb{R}$.*

Theorem 0.15.14 (Reciprocal Is a Multiplicative Inverse on $\mathbb{R}$). *If $\alpha \neq 0_{\mathbb{R}}$, then*

$$\alpha \cdot \alpha^{-1} = 1_{\mathbb{R}}.$$

Definition 0.15.15 (Division on $\mathbb{R}$). For $\beta \neq 0_{\mathbb{R}}$, define

$$\alpha / \beta := \alpha \cdot \beta^{-1}.$$

Theorem 0.15.16 (Distributivity on $\mathbb{R}$). *For all $\alpha, \beta, \gamma \in \mathbb{R}$,*

$$\alpha \cdot (\beta + \gamma) = \alpha \cdot \beta + \alpha \cdot \gamma.$$

0.16 Field and Ordered-Field Structure of $\mathbb{R}$

Theorem 0.16.1 (Nonzero $\mathbb{R}$ Is a Multiplicative Abelian Group). *The structure $(\mathbb{R} \setminus \{0_{\mathbb{R}}\}, \cdot, 1_{\mathbb{R}})$ is an abelian group.*

Theorem 0.16.2 ($\mathbb{R}$ Is a Field). *The structure $(\mathbb{R}, +, \cdot, 0_{\mathbb{R}}, 1_{\mathbb{R}})$ is a field.*

Corollary 0.16.3 (Field Laws Hold on $\mathbb{R}$). *Every theorem proved generically for fields holds in $\mathbb{R}$, including zero product, double negation, sign rules, cancellation, uniqueness of inverses, inverse laws, and exponent laws.*

Order Compatibility

Theorem 0.16.4 (Addition Preserves Order on $\mathbb{R}$). *For all $\alpha, \beta, \gamma \in \mathbb{R}$,*

$$\alpha < \beta \implies \alpha + \gamma < \beta + \gamma,$$

and the corresponding non-strict statement also holds.

Theorem 0.16.5 (Multiplication by Positives Preserves Order on $\mathbb{R}$). *For all $\alpha, \beta, \gamma \in \mathbb{R}$,*

$$0_{\mathbb{R}} < \gamma \wedge \alpha < \beta \implies \gamma \cdot \alpha < \gamma \cdot \beta.$$

Theorem 0.16.6 ($\mathbb{R}$ Is an Ordered Field). *The structure $(\mathbb{R}, +, \cdot, 0_{\mathbb{R}}, 1_{\mathbb{R}}, \leq)$ is an ordered field.*

Corollary 0.16.7 (Ordered-Field Laws Hold on $\mathbb{R}$). *Every theorem proved generically for ordered fields holds in $\mathbb{R}$.*

Theorem 0.16.8 ($\mathbb{R}$ Is a Complete Ordered Field). *The ordered field $\mathbb{R}$ satisfies the least-upper-bound property.*

0.17 Embedding $\mathbb{Q}$ into $\mathbb{R}$

Definition 0.17.1 (Embedding of $\mathbb{Q}$ into $\mathbb{R}$). Define $\iota : \mathbb{Q} \rightarrow \mathbb{R}$ by

$$\iota(q) := q^*.$$

Theorem 0.17.2 (Embedding Preserves Zero and One). *The embedding satisfies*

$$\iota(0_{\mathbb{Q}}) = 0_{\mathbb{R}} \quad \text{and} \quad \iota(1_{\mathbb{Q}}) = 1_{\mathbb{R}}.$$

Theorem 0.17.3 (Embedding $\mathbb{Q}$ into $\mathbb{R}$ Is Injective). *For all $p, q \in \mathbb{Q}$,*

$$\iota(p) = \iota(q) \implies p = q.$$

Theorem 0.17.4 (Embedding Preserves Addition and Multiplication). *For all $p, q \in \mathbb{Q}$,*

$$\iota(p + q) = \iota(p) + \iota(q) \quad \text{and} \quad \iota(p \cdot q) = \iota(p) \cdot \iota(q).$$

Theorem 0.17.5 (Embedding Preserves Order). *For all $p, q \in \mathbb{Q}$,*

$$p \leq q \iff \iota(p) \leq \iota(q).$$

Theorem 0.17.6 ($\mathbb{Q}$ Embeds into $\mathbb{R}$ as an Ordered Field). *The map ι is an injective order-preserving field homomorphism.*

Theorem 0.17.7 (Density of $\mathbb{Q}$ in $\mathbb{R}$). *For all $\alpha, \beta \in \mathbb{R}$,*

$$\alpha < \beta \implies \exists q \in \mathbb{Q} (\alpha < \iota(q) < \beta).$$

Theorem 0.17.8 (Archimedean Property of $\mathbb{R}$). *For $\alpha, \beta \in \mathbb{R}$,*

$$0_{\mathbb{R}} < \alpha \implies \exists n \in \mathbb{N} (\beta < \iota(n) \cdot \alpha).$$

0.18 The Gap Filled

Definition 0.18.1 (The Square-Root-of-Two Cut). Define

$$\sqrt{2} := \{r \in \mathbb{Q} : r < 0_{\mathbb{Q}} \vee r \cdot r < 2_{\mathbb{Q}}\}.$$

Lemma 0.18.2 (The Square-Root-of-Two Cut Is a Cut). *The set $\sqrt{2}$ is a Dedekind cut.*

Theorem 0.18.3 (Its Square Is Two). *In $\mathbb{R}$,*

$$\sqrt{2} \cdot \sqrt{2} = 2_{\mathbb{R}},$$

where $2_{\mathbb{R}} := \iota(2_{\mathbb{Q}})$.

Corollary 0.18.4 ($\mathbb{R}$ Contains the Supremum $\mathbb{Q}$ Was Missing). *Let*

$$S = \{q \in \mathbb{Q} : 0_{\mathbb{Q}} < q \wedge q \cdot q < 2_{\mathbb{Q}}\}.$$

Then

$$\sqrt{2} = \sup \iota(S).$$

Thus the bounded-above rational set with no rational supremum has a supremum in $\mathbb{R}$.

0.19 Characterization of $\mathbb{R}$

Definition 0.19.1 (Complete Ordered Field). A *complete ordered field* is an ordered field satisfying the least-upper-bound property.

Theorem 0.19.2 (Uniqueness of the Complete Ordered Field). *Any two complete ordered fields are isomorphic by a unique order-isomorphism.*

Corollary 0.19.3 ($\mathbb{R}$ Is the Real Numbers). *The field $\mathbb{R}$ is determined up to unique order-isomorphism by being a complete ordered field.*

0.20 Transition and Forward

Proofs

Capstone

Complex Numbers

0.21 Construction of $\mathbb{C}$

Complex Prepairs

Definition 0.21.1 (Complex Prepair). A *complex prepair* is an ordered pair $(a, b) \in \mathbb{R} \times \mathbb{R}$, read informally as the expression $a + bi$.

Definition 0.21.2 (Complex Prepair Set). The complex prepair set is

$$\text{Pre } \mathbb{C} := \mathbb{R} \times \mathbb{R}.$$

Coordinate Equivalence

Definition 0.21.3 (Complex Coordinate Equivalence). For $(a, b), (c, d) \in \text{Pre } \mathbb{C}$, define

$$(a, b) \sim_{\mathbb{C}} (c, d) \quad :\Longleftrightarrow \quad a = c \text{ and } b = d.$$

Lemma 0.21.4 (Complex Coordinate Equivalence Is Reflexive). *For every $(a, b) \in \text{Pre } \mathbb{C}$,*

$$(a, b) \sim_{\mathbb{C}} (a, b).$$

Lemma 0.21.5 (Complex Coordinate Equivalence Is Symmetric). *For all prepairs $(a, b), (c, d)$,*

$$(a, b) \sim_{\mathbb{C}} (c, d) \implies (c, d) \sim_{\mathbb{C}} (a, b).$$

Lemma 0.21.6 (Complex Coordinate Equivalence Is Transitive). *For all prepairs $(a, b), (c, d), (e, f)$,*

$$(a, b) \sim_{\mathbb{C}} (c, d) \wedge (c, d) \sim_{\mathbb{C}} (e, f) \implies (a, b) \sim_{\mathbb{C}} (e, f).$$

Theorem 0.21.7 (Complex Coordinate Equivalence Is an Equivalence Relation). *The relation $\sim_{\mathbb{C}}$ is an equivalence relation on $\text{Pre } \mathbb{C}$.*

Definition of $\mathbb{C}$

Definition 0.21.8 (Complex Numbers). The complex numbers are the quotient

$$\mathbb{C} := \text{Pre } \mathbb{C} / \sim_{\mathbb{C}}.$$

Definition 0.21.9 (Complex Class Notation). The notation $[a, b]_{\mathbb{C}}$ denotes the $\sim_{\mathbb{C}}$ -equivalence class of the prepair (a, b) .

Definition 0.21.10 (Zero, One, and the Imaginary Unit in $\mathbb{C}$). Define

$$0_{\mathbb{C}} := [0, 0]_{\mathbb{C}}, \quad 1_{\mathbb{C}} := [1, 0]_{\mathbb{C}}, \quad i := [0, 1]_{\mathbb{C}}.$$

0.22 Operations on $\mathbb{C}$ -Classes

Addition on $\mathbb{C}$ -Classes

Definition 0.22.1 (Addition on $\mathbb{C}$ -Classes). Define addition on representatives by

$$[a, b]_{\mathbb{C}} + [c, d]_{\mathbb{C}} := [a + c, b + d]_{\mathbb{C}}.$$

Lemma 0.22.2 (Addition Respects Complex Coordinate Equivalence). *If $[a, b]_{\mathbb{C}} = [a', b']_{\mathbb{C}}$ and $[c, d]_{\mathbb{C}} = [c', d']_{\mathbb{C}}$, then*

$$[a, b]_{\mathbb{C}} + [c, d]_{\mathbb{C}} = [a', b']_{\mathbb{C}} + [c', d']_{\mathbb{C}}.$$

Theorem 0.22.3 (Addition on $\mathbb{C}$ Is Well-Defined). *Addition determines a well-defined binary operation*

$$\mathbb{C} \times \mathbb{C} \rightarrow \mathbb{C}.$$

Negation and Subtraction

Definition 0.22.4 (Negation on $\mathbb{C}$ -Classes). Define negation by

$$-[a, b]_{\mathbb{C}} := [-a, -b]_{\mathbb{C}}.$$

Lemma 0.22.5 (Negation Respects Complex Coordinate Equivalence). *If $[a, b]_{\mathbb{C}} = [a', b']_{\mathbb{C}}$, then*

$$-[a, b]_{\mathbb{C}} = -[a', b']_{\mathbb{C}}.$$

Theorem 0.22.6 (Negation on $\mathbb{C}$ Is Well-Defined). *Negation determines a well-defined unary operation $\mathbb{C} \rightarrow \mathbb{C}$.*

Definition 0.22.7 (Subtraction on $\mathbb{C}$). For $z, w \in \mathbb{C}$, define

$$z - w := z + (-w).$$

Theorem 0.22.8 (Subtraction on $\mathbb{C}$ Is Well-Defined). *Subtraction determines a well-defined binary operation $\mathbb{C} \times \mathbb{C} \rightarrow \mathbb{C}$.*

Multiplication on $\mathbb{C}$ -Classes

Definition 0.22.9 (Multiplication on $\mathbb{C}$ -Classes). Define multiplication on representatives by

$$[a, b]_{\mathbb{C}} \cdot [c, d]_{\mathbb{C}} := [ac - bd, ad + bc]_{\mathbb{C}}.$$

Lemma 0.22.10 (Multiplication Respects Complex Coordinate Equivalence). *If $[a, b]_{\mathbb{C}} = [a', b']_{\mathbb{C}}$ and $[c, d]_{\mathbb{C}} = [c', d']_{\mathbb{C}}$, then*

$$[a, b]_{\mathbb{C}} \cdot [c, d]_{\mathbb{C}} = [a', b']_{\mathbb{C}} \cdot [c', d']_{\mathbb{C}}.$$

Theorem 0.22.11 (Multiplication on $\mathbb{C}$ Is Well-Defined). *Multiplication determines a well-defined binary operation*

$$\mathbb{C} \times \mathbb{C} \rightarrow \mathbb{C}.$$

Conjugation and Inverses

Definition 0.22.12 (Complex Conjugation). For $z = [a, b]_{\mathbb{C}}$, define its *complex conjugate* by

$$\bar{z} := [a, -b]_{\mathbb{C}}.$$

Theorem 0.22.13 (Complex Conjugation Is Well-Defined). *Complex conjugation determines a well-defined unary operation $\mathbb{C} \rightarrow \mathbb{C}$.*

Lemma 0.22.14 (Product with the Conjugate Is Real and Nonnegative). *For $z = [a, b]_{\mathbb{C}}$,*

$$z\bar{z} = [a^2 + b^2, 0]_{\mathbb{C}}.$$

Moreover, if $z \neq 0_{\mathbb{C}}$, then $a^2 + b^2 \neq 0$.

Definition 0.22.15 (Multiplicative Inverse in $\mathbb{C}$). *For $z = [a, b]_{\mathbb{C}} \neq 0_{\mathbb{C}}$, define*

$$z^{-1} := \left[\frac{a}{a^2 + b^2}, \frac{-b}{a^2 + b^2} \right]_{\mathbb{C}}.$$

Theorem 0.22.16 (Inversion on $\mathbb{C}^{\times}$ Is Well-Defined). *Inversion determines a well-defined operation on the nonzero complex numbers.*

0.23 Field Structure of $\mathbb{C}$

Additive Structure

Theorem 0.23.1 (Addition on $\mathbb{C}$ Is Associative). *For all $z, w, u \in \mathbb{C}$,*

$$(z + w) + u = z + (w + u).$$

Theorem 0.23.2 (Addition on $\mathbb{C}$ Is Commutative). *For all $z, w \in \mathbb{C}$,*

$$z + w = w + z.$$

Theorem 0.23.3 (Zero Is an Additive Identity on $\mathbb{C}$). *For every $z \in \mathbb{C}$,*

$$0_{\mathbb{C}} + z = z \quad \text{and} \quad z + 0_{\mathbb{C}} = z.$$

Theorem 0.23.4 (Every Complex Number Has an Additive Inverse). *For every $z \in \mathbb{C}$,*

$$z + (-z) = 0_{\mathbb{C}} \quad \text{and} \quad (-z) + z = 0_{\mathbb{C}}.$$

Multiplicative Structure

Theorem 0.23.5 (Multiplication on $\mathbb{C}$ Is Associative). *For all $z, w, u \in \mathbb{C}$,*

$$(zw)u = z(wu).$$

Theorem 0.23.6 (Multiplication on $\mathbb{C}$ Is Commutative). *For all $z, w \in \mathbb{C}$,*

$$zw = wz.$$

Theorem 0.23.7 (One Is a Multiplicative Identity on $\mathbb{C}$). *For every $z \in \mathbb{C}$,*

$$1_{\mathbb{C}}z = z \quad \text{and} \quad z1_{\mathbb{C}} = z.$$

Theorem 0.23.8 (Every Nonzero Complex Number Has a Multiplicative Inverse). *For every $z \in \mathbb{C}$ with $z \neq 0_{\mathbb{C}}$,*

$$zz^{-1} = 1_{\mathbb{C}} \quad \text{and} \quad z^{-1}z = 1_{\mathbb{C}}.$$

Distributivity and the Field Theorem

Theorem 0.23.9 (Multiplication Distributes over Addition on $\mathbb{C}$). *For all $z, w, u \in \mathbb{C}$,*

$$z(w + u) = zw + zu \quad \text{and} \quad (w + u)z = wz + uz.$$

Theorem 0.23.10 ($\mathbb{C}$ Is a Field). *With the operations defined above,*

$$(\mathbb{C}, +, \cdot, 0_{\mathbb{C}}, 1_{\mathbb{C}})$$

is a field.

The Imaginary Unit

Theorem 0.23.11 (The Imaginary Unit Squares to Negative One). *In $\mathbb{C}$,*

$$i^2 = -1_{\mathbb{C}}.$$

Theorem 0.23.12 ($\mathbb{C}$ Is Not an Ordered Field). *There is no order relation on $\mathbb{C}$ making $\mathbb{C}$ an ordered field and extending the usual order on $\mathbb{R}$.*

0.24 Embedding $\mathbb{R}$ into $\mathbb{C}$

The Real Axis

Definition 0.24.1 (Canonical Embedding of $\mathbb{R}$ into $\mathbb{C}$). Define

$$\iota : \mathbb{R} \rightarrow \mathbb{C}, \quad \iota(a) := [a, 0]_{\mathbb{C}}.$$

Theorem 0.24.2 (The Real Embedding into $\mathbb{C}$ Is Injective). *If $\iota(a) = \iota(b)$, then $a = b$.*

Theorem 0.24.3 (The Real Embedding Preserves Addition and Multiplication). *For all $a, b \in \mathbb{R}$,*

$$\iota(a + b) = \iota(a) + \iota(b), \quad \iota(ab) = \iota(a)\iota(b).$$

Theorem 0.24.4 (The Real Embedding Preserves Zero and One).

$$\iota(0) = 0_{\mathbb{C}}, \quad \iota(1) = 1_{\mathbb{C}}.$$

Real and Imaginary Parts

Definition 0.24.5 (Real and Imaginary Parts). For $z = [a, b]_{\mathbb{C}}$, define

$$\operatorname{Re}(z) := a, \quad \operatorname{Im}(z) := b.$$

Theorem 0.24.6 (Real and Imaginary Parts Are Well-Defined). *The functions*

$$\operatorname{Re}, \operatorname{Im} : \mathbb{C} \rightarrow \mathbb{R}$$

are well-defined.

Proofs

Capstone

Number Lines and Intervals

0.25 Number Lines

Definition 0.25.1 (Number Line). A *number line* for a linearly ordered system $(A, <_A)$ is a linearly ordered set of points L together with an order isomorphism $\lambda : A \rightarrow L$, so that for all $a, a' \in A$,

$$a <_A a' \iff \lambda(a) \text{ lies to the left of } \lambda(a').$$

0.26 Intervals on a Number Line

Definition 0.26.1 (Open Interval). Let $a, b \in \mathbb{R}$ with $a \leq b$. The *open interval* with endpoints a and b is

$$(a, b) = \{x \in \mathbb{R} : a < x < b\}.$$

Definition 0.26.2 (Closed Interval). Let $a, b \in \mathbb{R}$ with $a \leq b$. The *closed interval* with endpoints a and b is

$$[a, b] = \{x \in \mathbb{R} : a \leq x \leq b\}.$$

Definition 0.26.3 (Left-Half-Open Interval). Let $a, b \in \mathbb{R}$ with $a \leq b$. The *left-half-open interval* with endpoints a and b is

$$(a, b] = \{x \in \mathbb{R} : a < x \leq b\}.$$

Definition 0.26.4 (Right-Half-Open Interval). Let $a, b \in \mathbb{R}$ with $a \leq b$. The *right-half-open interval* with endpoints a and b is

$$[a, b) = \{x \in \mathbb{R} : a \leq x < b\}.$$

Definition 0.26.5 (Ray). Let $a \in \mathbb{R}$. The four *rays* determined by a are

$$(a, \infty) = \{x : x > a\}, [a, \infty) = \{x : x \geq a\}, (-\infty, a) = \{x : x < a\}, (-\infty, a] = \{x : x \leq a\}.$$

Proofs

Capstone

Embedding the Number Systems

0.27 Embeddings as Structure Preserving Maps

Definition 0.27.1 (Structure-Preserving Map). Let A and B be ordered arithmetic systems with a specified common source structure: selected constants, selected operations, and a strict order. A map $\varphi : A \rightarrow B$ is *structure-preserving* exactly when it preserves each selected constant and, for every selected binary operation $\star$ and all $a, a' \in A$,

$$\varphi(a \star_A a') = \varphi(a) \star_B \varphi(a'),$$

and $a <_A a' \implies \varphi(a) <_B \varphi(a')$.

Definition 0.27.2 (Embedding of Number Systems). Let A and B be ordered arithmetic systems. A map $\varphi : A \rightarrow B$ is an *embedding* exactly when φ is injective and structure-preserving and is an order embedding.

Definition 0.27.3 (Image of an Embedding). Let $\varphi : A \rightarrow B$ be an embedding. The *image* of φ is the set $\varphi(A) = \{ \varphi(a) : a \in A \} \subseteq B$, equipped with the selected constants, operations, and order of B restricted to $\varphi(A)$.

Theorem 0.27.4 (Embeddings Identify a System with Its Image). *Let $\varphi : A \rightarrow B$ be an embedding of ordered arithmetic systems relative to a selected source structure. Then the corestriction $\varphi : A \rightarrow \varphi(A)$ is an isomorphism onto its image: it is a bijection that preserves and reflects the selected constants, operations, and order. Consequently A may be identified with the corresponding substructure $\varphi(A) \subseteq B$. Go to proof.*

0.28 Embedding $\mathbb{N}$ Into $\mathbb{W}$

Definition 0.28.1 (Embedding $\mathbb{N}$ into $\mathbb{W}$). The *canonical embedding* $\iota : \mathbb{N} \rightarrow \mathbb{W}$ is defined by

$$\iota(n) = n \quad (\text{the copy of each natural number inside the whole numbers}).$$

Theorem 0.28.2 ($\mathbb{N}$ Embeds in $\mathbb{W}$). *The canonical map $\iota : \mathbb{N} \rightarrow \mathbb{W}$ identifies $\mathbb{N}$ with the positive part $\mathbb{W} \setminus \{0\}$ and preserves successor, 1, addition, multiplication, and order. Go to proof.*

0.29 Embedding $\mathbb{W}$ Into $\mathbb{Z}$

Definition 0.29.1 (Embedding $\mathbb{W}$ into $\mathbb{Z}$). The *canonical embedding* $\iota : \mathbb{W} \rightarrow \mathbb{Z}$ is defined by

$$\iota(w) = [(w, 0)] \quad (\text{the integer represented by } w \text{ and } 0).$$

Theorem 0.29.2 ($\mathbb{W}$ Embeds in $\mathbb{Z}$). *The canonical map $\iota : \mathbb{W} \rightarrow \mathbb{Z}$ is an embedding of ordered arithmetic systems: it is injective, preserves 0, 1, addition, and multiplication, and reflects the order. Go to proof.*

0.30 Embedding $\mathbb{Z}$ Into $\mathbb{Q}$

Definition 0.30.1 (Embedding $\mathbb{Z}$ into $\mathbb{Q}$). The *canonical embedding* $\iota : \mathbb{Z} \rightarrow \mathbb{Q}$ is defined by

$$\iota(z) = [(z, 1)] \quad (\text{the rational represented by } z/1).$$

Theorem 0.30.2 ($\mathbb{Z}$ Embeds in $\mathbb{Q}$). *The canonical map $\iota : \mathbb{Z} \rightarrow \mathbb{Q}$ is an embedding of ordered arithmetic systems: it is injective, preserves 0, 1, addition, and multiplication, and reflects the order. Go to proof.*

0.31 Embedding $\mathbb{Q}$ Into $\mathbb{R}$

Definition 0.31.1 (Embedding $\mathbb{Q}$ into $\mathbb{R}$). The *canonical embedding* $\iota : \mathbb{Q} \rightarrow \mathbb{R}$ is defined by

$$\iota(q) = q^* \quad (\text{the real number determined by } q).$$

Theorem 0.31.2 ($\mathbb{Q}$ Embeds in $\mathbb{R}$). *The canonical map $\iota : \mathbb{Q} \rightarrow \mathbb{R}$ is an embedding of ordered arithmetic systems: it is injective, preserves 0, 1, addition, and multiplication, and reflects the order. Go to proof.*

0.32 Compatibility of Arithmetic and Order

Theorem 0.32.1 (Compatibility of the Embedding Chain). *Let $\iota_1 : \mathbb{N} \rightarrow \mathbb{W}$, $\iota_2 : \mathbb{W} \rightarrow \mathbb{Z}$, $\iota_3 : \mathbb{Z} \rightarrow \mathbb{Q}$, and $\iota_4 : \mathbb{Q} \rightarrow \mathbb{R}$ be the canonical embeddings. For every $n \in \mathbb{N}$ the composite $(\iota_4 \circ \iota_3 \circ \iota_2 \circ \iota_1)(n)$ is the real number identified with n , and the composite preserves the arithmetic and order carried by the source system. Hence arithmetic computed in any system agrees with arithmetic on its image in $\mathbb{R}$. Go to proof.*

0.33 One Chain Many Structures

Proofs

Capstone

Summary: The Number Systems

0.34 Axiom Systems for $\mathbb{N}$, $\mathbb{W}$, $\mathbb{Z}$, $\mathbb{Q}$, and $\mathbb{R}$

0.35 Number Systems Summary

The Natural Numbers $\mathbb{N}$

The Whole Numbers $\mathbb{W}$

The Integers $\mathbb{Z}$

Derived Properties of $\mathbb{Z}$ (from the Ring Axioms)

Structural Comparison: $\mathbb{N}$, $\mathbb{W}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$

Proofs

Capstone